

Metabook: A System to Automatically Generate Interactive AR Storybooks to Improve Children's Reading Interest

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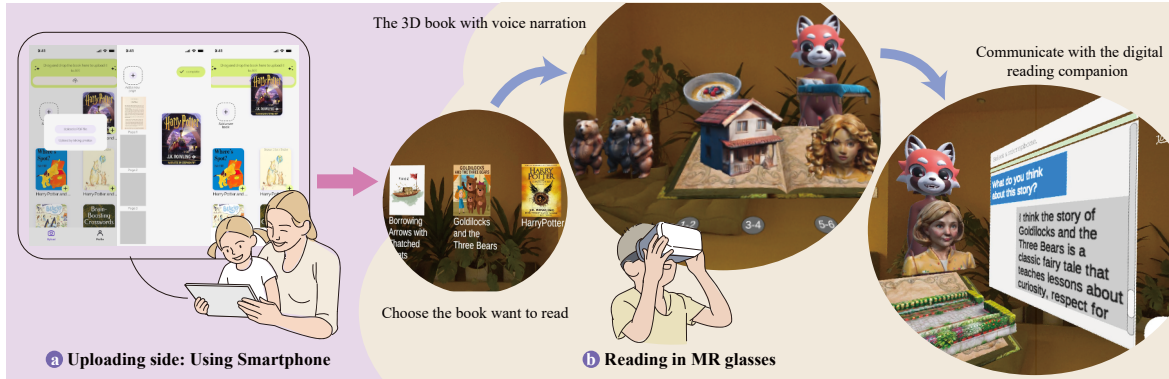


Figure 1: Overview of Metabook. Metabook is a system that automatically generates interactive AR storybooks. (a) Users upload books in PDF or image format using their smartphones. (b) Users enjoy an immersive reading experience of the 3D AR version of the uploaded book and communicate with the digital reading companion through MR glasses

ABSTRACT

We propose Metabook, a system to automatically generate interactive AR storybooks to improve children's reading interest. Metabook introduces a story-to-3D-book generation scheme and a 3D avatar that combines multiple AI models as a reading companion. Our user study shows that Metabook can significantly increase children's interest in reading. Teachers acknowledged Metabook's effectiveness in enhancing reading enthusiasm by connecting verbal and visual thinking, expressing high expectations for its future potential in education.

Index Terms: Augment reality, generative AI, children reading tool, AR book.

1 INTRODUCTION

Reading is important for children to acquire knowledge, enhance cognitive abilities, and improve language skills. However, current reading methods either offer limited visual presentation, making them less interesting to children, or lack channels for children to share insights and ask questions during reading. AR/VR books provide rich visual cues that address the issue of children's lack of interest in reading [1, 2], but the high production costs and need for professional expertise limit the volume of AR/VR books and children's choices.

We propose Metabook, a system to automatically generate interactive AR storybooks. We propose a **story to 3D book generation pipeline**, enabling even novice users without prior modeling

experience to swiftly create 3D storybooks via simple uploading operations on their smartphones. Our system also includes a digital reading companion, powered by large language models as well as action and speech generation models to enable communication. Our user study indicates that Metabook can significantly increase children's interest and willingness to read. Teachers also recognized Metabook's role in enhancing children's reading enthusiasm and building connections between verbal and visual thinking.

2 THE METABOOK SYSTEM

Our system is divided into three parts: the mobile front-end where users can upload physical books, the server back-end used to generate 3D versions of the books, and the MR glasses through which users can download 3D books from the server and enjoy the AR 3D reading experience. The architecture is shown in figure 2.

2.1 Story to 3D Book Generation

Mainstream 2D diffusion-based 3D generation methods are not suitable for producing 3D books for children because they struggle with prioritization when dealing with lengthy texts, and sometimes produce models that do not correspond to reality or text, which may mislead children.

Therefore, we present a story-to-3D-book generation scheme. The scheme includes 4 parts: a keyword scheme that combines character extraction and object selection to overcome the difficulties of long-story sentence prioritization. A TripoSR [5] for 3D models generation by calling API.

We also include a CLIP [3] model to assess the alignment between generated models and text. We denote the frontal view image and side view image of the generated model as I_1 and I_2 respectively. K represents the input text corresponding to each object generated and the threshold is C . The formula for calculating the matching degree S is as follows:

$$S = \min(\text{CLIP}(I_1, K), (\text{CLIP}(I_2, K))) \quad (1)$$

While $S < C$, the model will be regenerated. If the model still fails the rationality detection after the fifth attempt, the input will be

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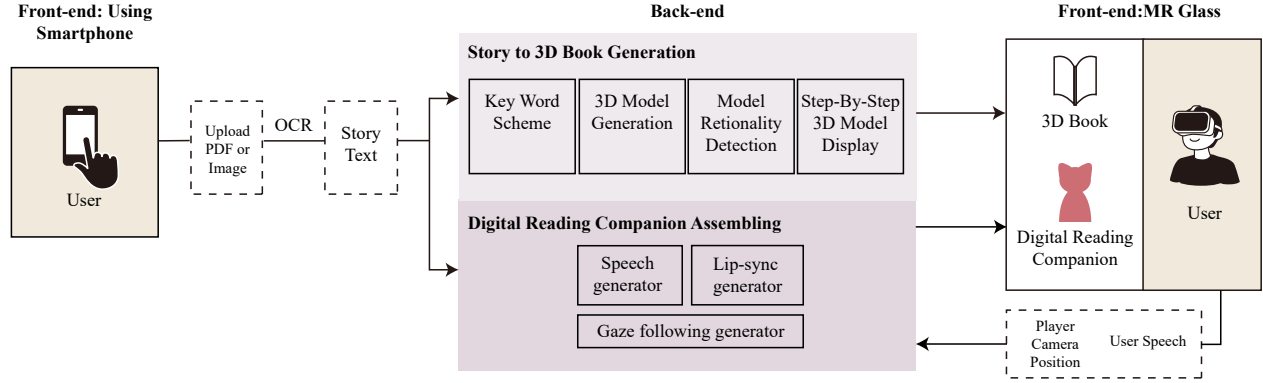


Figure 2: The architecture of the Metabook system, which consists of three parts: the front-end uploading part using a smartphone, the back-end Metabook production part, and the reading part within MR glasses. Users can upload a book in PDF or image format, and then enjoy an immersive 3D AR book reading experience accompanied by an interactive digital reading companion.

discarded. We also introduce a module controlling the model appearing time. The formula for appearing time T is as follows:

$$T = \lceil P_K / r \times 5 \rceil \quad (2)$$

T represents the appearance time of the 3D model, and P_K represents how many words are before the character/object keywords. And r represents the number of words read every 5 seconds.

2.2 Digital Reading Companions

To enhance children's enthusiasm for reading books, we use OVR-LipSync to generate the 3D avatar's lip movement synchronization and Eye Animator to ensure that the 3D avatar's gaze continuously follows the player. We connect GPT-4 with the avatar to create reasonable responses.

3 RESULTS

We recruited 44 children to participate in the study. Among them, there were 20 boys and 24 girls, with an average age of 11.1 years old. All participants were between 10 and 12 years old.

We divided children into two equal groups, A (A1 to A22) and B (B1 to B22), based on similar age. We selected two stories of similar difficulty and length from the same chapter of a higher-grade Chinese textbook. Both groups first read story 1¹ on printed paper. Then, students in Group A completed reading story 2² in MR without having a conversation with the digital reading companion. Students in Group B have a conversation with the digital reading companion after reading story 2.

We adopted adapted Smileyometer [4, 6] and Again-Again table [4, 6] to measure children's reading interest. We used a 1–5 Likert scale, from “very boring” to “very interesting.” In the Again-Again Table, 1 represents “not willing,” 2 represents “maybe willing” and 3 represents “willing.”

Overall, children's interest in reading significantly increased through the use of Metabook. In the Smileyometer ratings for the level of interest, Group A gave scores of 3/5 and 4.5/5 for paper reading and Metabook reading ($p = 0.00003$). Group B's scores for paper reading and Metabook reading were 3.6/5 and 4.6/5. ($p = 0.00025$). Their willingness to read another story also increased significantly. In the Again-Again Table, Group A scored 2/3, and Group B scored 2.2/3 for their willingness to read another story with paper books, while both groups gave a strong score

of 3/3 for their willingness to read another story using Metabook ($p = 0.0002$ for Group A and $p = 0.0005$ for Group B).

After experiencing Metabook, the teachers expressed their positive attitude towards Metabook. TS3 stated, “This storytelling method with cartoon characters can spark children's curiosity about what will happen next.” TS6 suggested, “It could be used as an assistant, with the digital character leading group discussions.”

4 CONCLUSION AND DISCUSSION

We proposed Metabook in this paper, a system to generate interactive AR storybooks to improve children's reading interest. Our user study shows that children's interest and willingness to read another story have been raised significantly. We see the potential of this system as a future after-school reading tool for children. We hope to further improve the system and conduct a more comprehensive evaluation.

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¹Wu Song Fights the Tiger

²Borrowing Arrows with Thatched Boats